Example 7.70

Sketch the curve $y = f(x) = x^3 - 6x - 9$.

Solution

Factorising the given function, we have

$$y = f(x) = (x-3)(x^2+3x+3).$$

(1) The domain and the range of the given function $f(x)$ are the entire real line.

(2) Putting $y = 0$, we get the $x = 3$. The other two roots are imaginary. Therefore, the x-intercept is $(3,0)$. Putting $x = 0$, we get $y = -9$. Therefore, the y-intercept is $(0,-9)$.

(3) $f'(x) = 3(x^2-2)$ and hence the critical points of the curve occur at $x = \pm\sqrt{2}$.

(4) $f''(x) = 6x$. Therefore at $x = \sqrt{2}$ the curve has a local minimum because $f''\left(\sqrt{2}\right) = 6\sqrt{2} > 0$. The local minimum is $f\left(\sqrt{2}\right) = -4\sqrt{2} - 9$. Similarly $x = -\sqrt{2}$ the curve has a local maximum because $f''\left(-\sqrt{2}\right) = -6\sqrt{2} < 0$. The local maximum is $f\left(-\sqrt{2}\right) = 4\sqrt{2} - 9$.

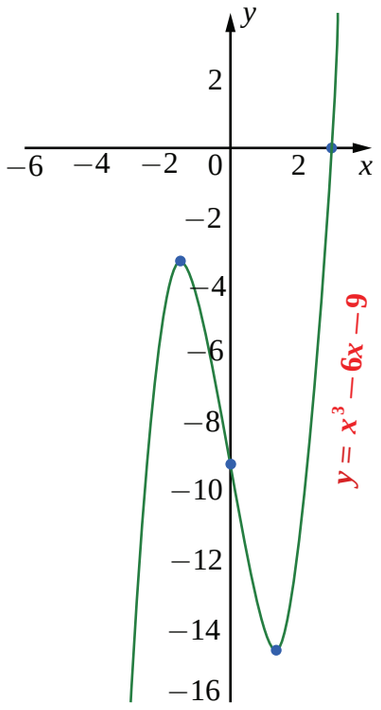


Fig.7.36

(5) Since $f''(x) = 6x > 0,\ \forall\, x > 0$ the function is concave upward in the positive real line. As $f''(x) = 6x < 0,\ \forall\, x < 0$ the function is concave downward in the negative real line.

(6) Since $f''(x) = 0$ at $x = 0$ and $f''(x)$ changes its sign when passing through $x = 0$. Therefore the point of inflection is $(0, f(0)) = (0,-9)$.

(7) The curve has no asymptotes.

The rough sketch of the curve is shown on the right side.

Example 7.71

Sketch the curve $y = \dfrac{x^2 - 3x}{(x-1)}$.

Solution

Factorising the given function we have,

$$y = f(x) = \frac{x(x-3)}{(x-1)}.$$

(1) The domain and the range of $f(x)$ are respectively $R \setminus \{1\}$ and the entire real line.

(2) Putting $y = 0$ we get the $x = 0, 3$. Therefore the x-intercept is $(3,0)$. Putting $x = 0$, we get $y = 0$. Therefore the curve passes through the origin.

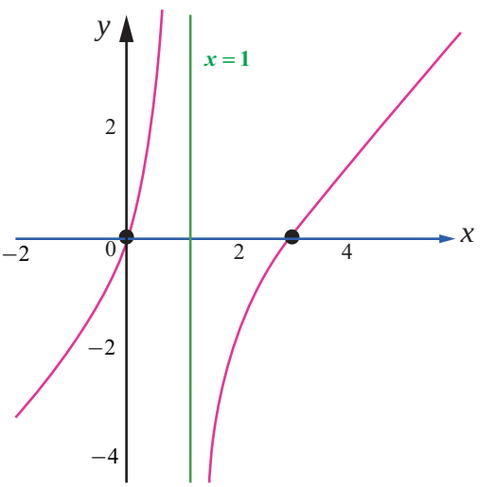


Fig.7.37

12th_Maths_Vol 2_EM_CH 7_Differential Calculus.indd 51 16-12-2022 16:28:36